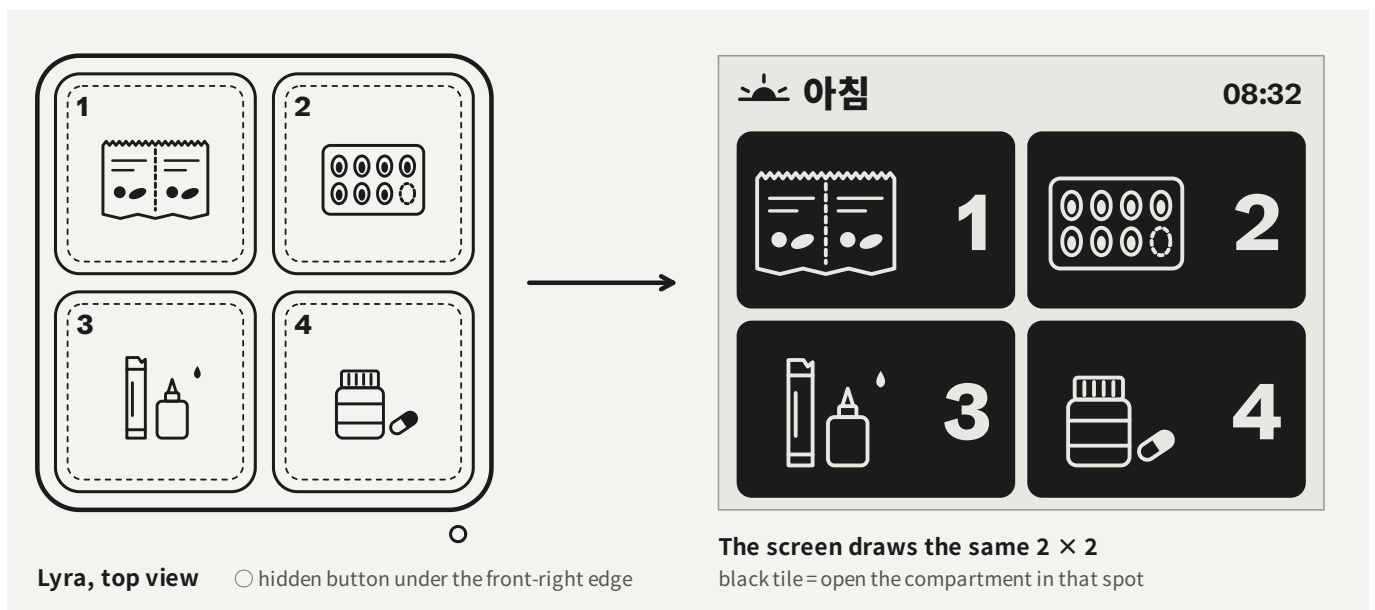


Lyra

A medication station that shows what to open, and knows when it was opened.



Brief for Incheon Techno Park

26 September 2026 · navdesignit

Logic, calculations, failure scenarios, every screen, AI, the link to Agvika OS, and the plan. Estimates are marked as estimates.

One-page summary

Lyra is a medication station for people who live alone. It has four compartments and a 4.2-inch e-ink screen. The screen shows which compartment to open now. Lyra records when each compartment was opened.

- **The problem.** Older people who live alone forget doses, take a dose twice, or skip doses when they go out. Often nobody notices until something goes wrong.
- **What Lyra does.** It lights up the compartment that is due. It records the time the compartment was opened and whether its weight went down.
- **What Lyra never claims.** It does not know which pill was taken, how many, or whether it was swallowed. It knows only the compartment, the time and the weight change.
- **Why it is new.** It works with Korean medicine as people receive it: printed sachet rolls, blister packs, sticks and bottles. There is nothing to refill into cups, no app and no camera.
- **Link to Agvika OS.** Lyra sends about 3 KB of routine events a day. Agvika OS uses them as a daily sign that the person is keeping their routine.
- **Evidence today.** The core firmware is written in C++ and tested in simulation. 200 random simulated days pass. 500 knocks on the device cause 0 false "opened" records.
- **Proposed ask.** A 20-home pilot in Incheon, lab support to build the first 30 units, and help with certification (details in the last section).

The one-line promise to the user: *Lyra shows you what to open now. If a dose time is missed, it tells your family once.*

The problem in Korea

More Koreans live alone in old age every year, and most of them take many medicines. When they miss a dose, often nobody is there to notice.

Fact	Number	Source
Share of Koreans aged 65 or over	20% in December 2024 (10.24 million people). Korea became a "super-aged" society.	Korea Times , MOIS data
Older households with only one person	37.8% in 2025, the most common household type among older people	Statistics Korea , 2025 고령자 통계
Older Koreans with polypharmacy (6 or more medicines)	86.4%	PLOS ONE , 2014

These figures come from search result summaries. The pages could not be opened from this workspace, so please check them before sending.

Four everyday failures Lyra targets:

1. **Forgetting a dose time.** The medicine is in a drawer, and nothing reminds the person at lunch.
2. **"Did I already take it?"** Without a record, people either skip the dose or take it twice.
3. **Going out.** The lunch sachet stays at home when the person leaves at 11:40.
4. **Family far away.** A son or daughter in Seoul finds out weeks later, at a hospital visit.

Why today's products do not fit Korean homes:

- Korean pharmacies hand out printed sachet rolls (약포지) already sorted by dose time. Pill-cup dispensers make people cut the sachets open and re-sort the pills.
- A real Korean routine is not only pills. It has blisters, red ginseng sticks, eye drops, ointment and supplement bottles.
- Imported dispensers are subscriptions. Hero costs \$99.99 upfront plus \$29.99 to \$44.99 a month. Livi costs \$130 upfront plus \$99 a month.
- Phone apps expect an older person to use an app every day. Lyra has its own screen, so there is no app for the user.

What Lyra is

Lyra is a tabletop box with four open compartments in a 2×2 square. Its 4.2-inch e-ink screen draws the same square, and each compartment has its own small scale and light. People put their medicine in as they receive it. They never repack it.

Compartment	What goes in	How people use it	Largest normal "use" (weight change)
1 · Sachets	Printed vinyl sachet roll from the pharmacy, or loose sachets	Tear off one sachet, put the roll back	3 g
2 · Blisters	Blister packs from the pharmacy or hospital	Press out a pill, put the pack back	1 g
3 · Packages	Red ginseng sticks, eye drops, ointment	Take a stick, or lift a tube, use it, put it back	13 g
4 · Bottles	Supplement bottles	Lift a bottle, shake out a capsule, put it back	2 g

The last column matters. If more weight than that leaves a compartment, Lyra knows the whole container was taken out, not just one dose.

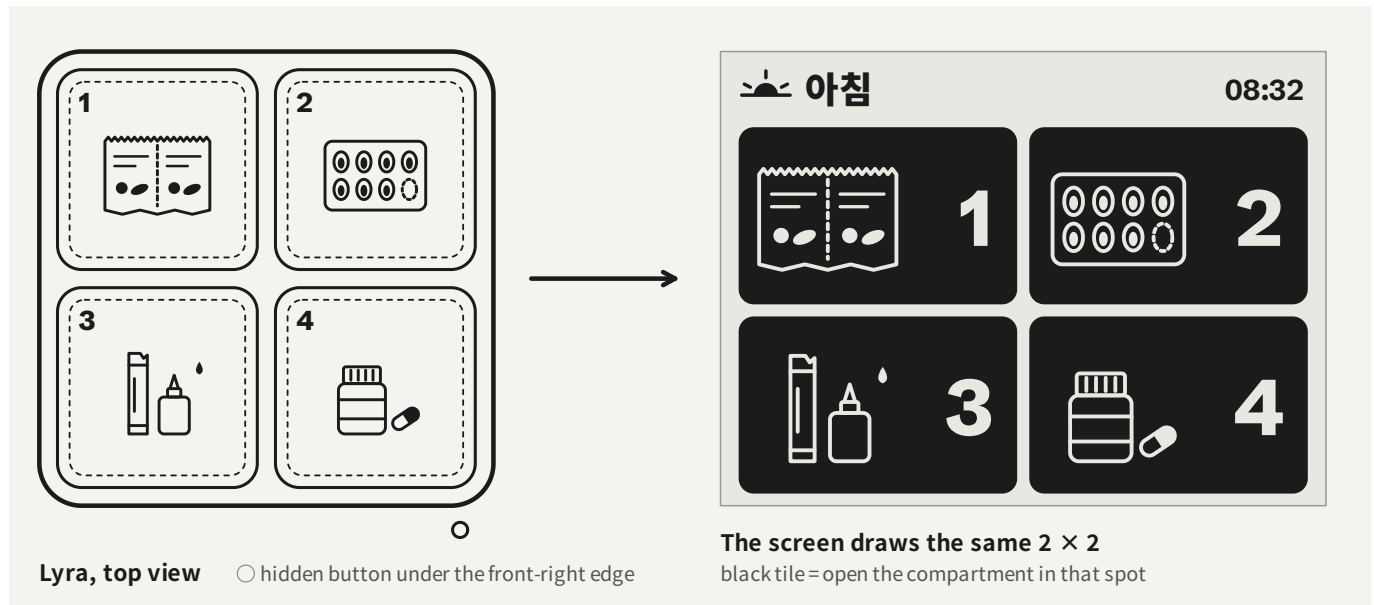
Parts inside (first prototype):

- ESP32-S3 controller with Wi-Fi
- 4.2-inch e-ink screen, 400×300 pixels, readable in daylight and using no power while it shows a still image
- 4 load cells with HX711 amplifiers, one scale per compartment, read 10 times a second
- 24 GHz radar (HLK-LD2410C) that tells if someone is in the room. It is not a camera and records no image.
- Temperature and humidity sensor (SHT40)
- Speaker for a soft chime and a short Korean voice line
- One small button hidden under the front-right edge, the only control. There is no knob.
- 4 lights, one under each compartment
- Real-time clock and a backup battery, so the schedule survives a power cut

How a person sets it up: a family member or pharmacist opens a web page on any phone and enters the dose times for each compartment. The older person never installs anything.

Every screen

The four compartments sit in a 2×2 square, and the screen draws the same square. Each tile sits exactly where its compartment sits, with a picture of what is inside. So the person needs no numbers and no reading.



On a tile	Meaning	What the person does
Black, white picture	Due now. Open the compartment in this spot.	Opens it
Outline, check mark, time	Opened at that time	Nothing
Dotted, grey picture	Not needed now	Nothing
Slash across	The dose time ended without an opening	Nothing. Family was told once.
Dashed, arrow down	A container is out. Put it back here.	Puts it back
Thick outline	Opened when it was not due	Nothing. The next time is shown.

The screen is 400×300 pixels, black and white. Grey is a 50% dot pattern, as a 1-bit e-ink panel shows it. A page change uses a full refresh (one short flash), and a tile flipping uses a quiet partial refresh.

What shows when someone comes near

The radar tells Lyra when a person is within about 1.5 m. The screen then shows the single most important thing, in this order:

1. A container has been out for 10 minutes: **put it back** (screen 9).
2. The person is leaving near a dose time: **take it with you** (screen 8).
3. A dose time is open: **the 2×2 map**, black = open (screens 4 to 6).
4. AI: a usual outing is coming: **pack the next dose** (screen 13).
5. AI: 3 days of sachets or fewer: **pharmacy day** (screen 12).
6. A dose time just ended with a compartment unopened: **slash, family told** (screen 10).
7. Nothing due: **the next dose time** and which compartments (screen 3).

When nobody is near, it shows the clock, weather and fine dust (screens 1, 2 and 11).

No knob. One small button is hidden under the front-right edge. A short press shows today (screen 15). Holding it for 3 seconds starts refill mode for the family or carer (screen 16).

Resting and near



1 · Resting, clean air (14:20). Clock, date, weather and temperature. Fine dust is "좋음" (good), 1 dot of 4. The bottom row shows the next dose time, 18:30, and a tiny 2 × 2 map: black squares for compartments 1, 2 and 4.



2 · Resting, very bad air (07:10). "매우나쁨" (very bad), 4 dots of 4, and "외출 시 마스크" (mask when you go out). This is the daily reason to look at Lyra.



3 · Someone comes near, nothing due (14:22). "다음 약" (next medicine): evening, 18:30, in 4 hours. The map shows the sachets, blisters and bottle that will be needed. The sticks and drops are grey because they are not needed then.

Dose time



4 · Morning dose (08:32). All four tiles are black, and the lights in all four compartments glow. A chime follows at +15 minutes and a voice line at +30.



5 · Two opened (08:42). The sachets and blisters were opened at 08:41 and 08:42. The sticks and the bottle are still black.

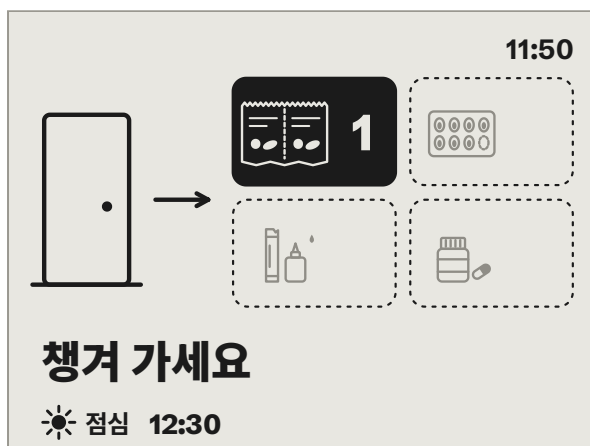


6 · All opened (08:45). Four checks with times. The lights go off and the screen returns to the clock.

Short notices



7 · Opened when not due (12:32). "지금은 아니에요" (not now). The bottle tile has a thick outline and shows when it is next due: evening, 18:30. There is no alarm.



8 · Leaving before lunch (11:50). A door, an arrow, and the sachet tile in black: "챙겨 가세요" (take it with you), lunch 12:30. It is said once per dose time.

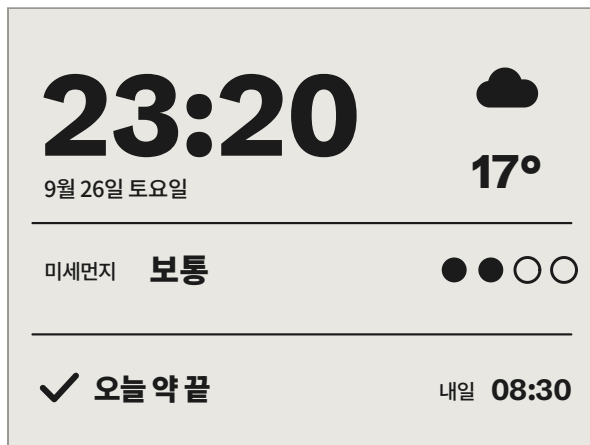


9 · Put it back (18:33). The bottle has been out for 10 minutes. Its tile is dashed with an arrow down, so the person sees exactly where it goes.

Missed and end of day



10 · Missed (21:31). Evening has ended. Sachets and bottle were opened at 18:35 and 18:36. The blister tile has a slash, and sticks were not needed. "지은님께 알렸어요" means Ji-eun (the daughter) was told.



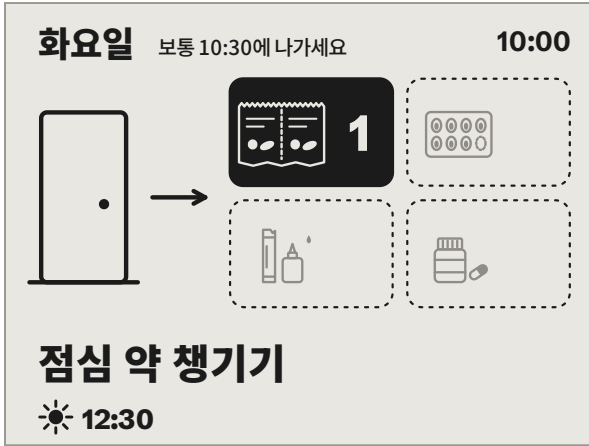
11 · All done (23:20). "오늘 약 끝" (today's medicine is done), and tomorrow starts at 08:30.

AI results on the screen

Yes, Lyra can show what it learns, as long as each result asks for one clear action. Three do:



12 · Sachets running out. "3일분 남았어요" (3 days left). Three black day squares (Sat, Sun, Mon) and a dotted one (Tue) show when the roll runs out. "월요일에 약국" means pharmacy on Monday. The calculation is in the AI section.

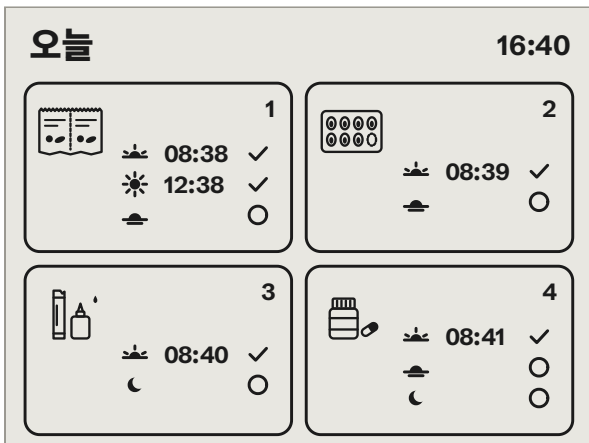


13 · A learned outing (Tuesday 10:00). Lyra has learned that the person usually leaves at 10:30 on Tuesdays and is back after lunch. So at 10:00 it says "점심 약 챙기기" (pack the lunch medicine) and shows the sachet tile.



14 · This week (Sunday morning). Each circle is one dose time on one day. A black check means everything needed was opened, and a slashed circle means one was missed. "67 / 70" counts the needed openings that were on time.

The hidden button



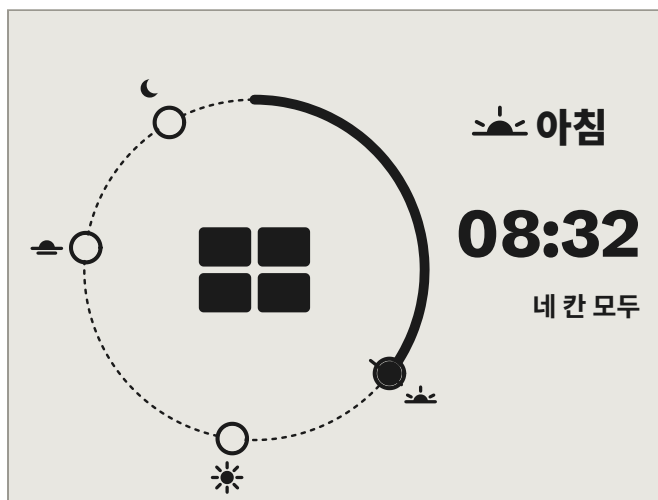
15 · Short press: today. Each tile lists its dose times: a time and a check if opened, an empty circle if still to come.



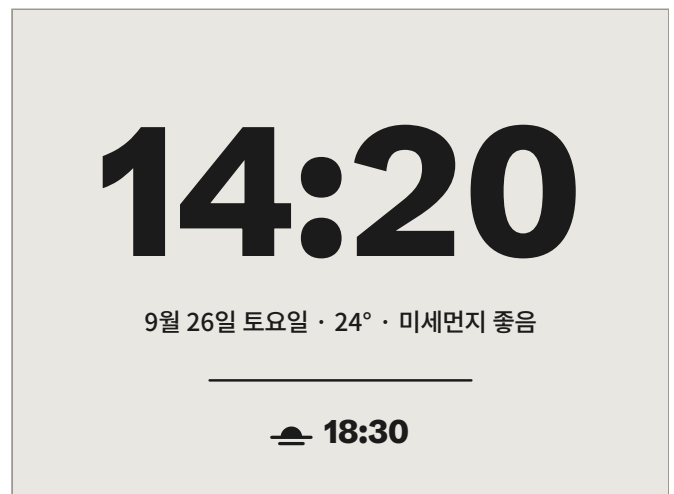
16 · Hold 3 seconds: refill mode. Compartments that gained weight show how much (+42 g for a new sachet roll, +118 g for a new bottle). Nothing is counted as an opening. One press ends it.

Design concepts compared

We tried three ways to show the same logic. The 2×2 map (A, all screens above) is the one we recommend.



B · Ring. The day is a 24-hour ring with the dose times as beads. It is beautiful and shows the whole day, but the compartments shrink to a small map in the middle.



C · One thing. One compartment at a time, as large as possible, with the next ones listed. It is the easiest to read from far away, but it adds steps when four compartments are due together.

Concept	Best at	Weak at	Verdict
A · Map	Showing what to open and where, at a glance	Showing the whole day	Recommended

Concept	Best at	Weak at	Verdict
B · Ring	Showing the day's rhythm	Showing which compartment	Resting screen option
C · One thing	Poor eyesight, reading from far away	Several compartments due at once	Large-text setting

Design files. Every screen and concept is in the repository as vector SVG (for Adobe Illustrator) and as PNG, in `lyra/design`. The UI kit (map, icons, tile states, rules, all screens and concepts) is one SVG artboard.

What we promise, and what we never claim

Lyra promises three facts about each compartment: that it was opened, when, and whether something came out. Everything it says to the person or the family is built from those three facts and the schedule.

What Lyra promises the user

1. At every dose time, it shows which compartments to open, following the schedule the family or pharmacist entered.
2. It records every opening: which compartment and the time, to the minute.
3. It records whether weight left the compartment ("something came out": yes or no).
4. It reminds you in steps, and only when you are home: a light first, then a chime at +15 minutes, then a voice at +30 minutes.
5. If a dose time ends and a needed compartment was not opened, it tells one family member once. It never sends a stream of alerts.
6. If you leave home less than 60 minutes before a dose time, it says "take it with you" once.
7. It keeps working without Wi-Fi, and it keeps the schedule through a power cut.
8. It never records sound, images or video.

What Lyra never claims

Lyra does not know	Why it cannot know	What protects the person instead
Which sachet was torn from a roll (morning, lunch or evening)	All sachets on a roll weigh about the same. Lyra only sees that about 1.5 g left compartment 1.	The dose time printed on each sachet by the pharmacy
Which item in a shared compartment was used (eye drops or ointment)	They share one scale	Different compartments for things that must not be mixed up
How many pills came out	One pill weighs 0.1 to 0.4 g, too close to the scale's noise to count	The blister pack itself shows the empty pockets
That the medicine was swallowed	No sensor can see this without a camera	The person's own habit, which Lyra supports
Who opened it	The scale cannot tell people apart	The log shows times, so the family can ask
Whether a dose is medically right	Lyra is not a medical device and gives no advice	The doctor and pharmacist

The words on screen and in family messages follow these limits. Lyra says "opened" and "not opened". It never says "took the medicine".

How Lyra calculates, step by step

Every decision Lyra makes comes from grams, seconds and minutes, and each one is shown below with real numbers. The noise figures are our estimates for the chosen load cells. They will be measured on the first prototypes.

Step 1. Read each scale 10 times a second

Each compartment's scale gives a reading every 0.1 s. Each reading wobbles a little, which we call the noise (σ).

Compartment	Noise per reading (σ)	Put-down repeatability (p)
1 · Sachets	0.02 g	0.02 g
2 · Blisters	0.015 g	0.02 g
3 · Packages	0.04 g	0.03 g
4 · Bottles	0.04 g	0.03 g

Step 2. Know when the scale is still

Lyra looks at the last 10 readings (1 second). If they span less than 6σ (at least 0.06 g), the scale is "settled", and the settled level is their average.

Example, compartment 1: $6 \times 0.02 \text{ g} = 0.12 \text{ g}$. Readings between 212.43 g and 212.49 g span 0.06 g, so the scale is settled at about 212.46 g.

Step 3. Know when a hand is in the compartment

A reading more than 8σ (at least 0.2 g) away from the settled level means a hand is pressing or lifting. For compartment 1 that is 0.2 g, and for compartments 3 and 4 it is 0.32 g. A finger resting on a sachet roll adds several grams, so it is easy to see.

Step 4. Know when the person has finished

An opening ends when the scale is settled and nothing has moved for 8 seconds. "Moved" means a hand, or a wobble larger than 15σ (at least 0.3 g), such as something being put down.

Step 5. Measure what came out

Net change = settled weight before the opening – settled weight after it.

Example, compartment 1: 212.46 g before, 210.94 g after. Net = 1.52 g, which is one sachet.

Step 6. Decide "something came out"

Both the before and after weights carry averaged noise ($\sigma/\sqrt{10}$) and put-down error (p). Lyra says "something came out" only when the net change is more than 4 times their combined error:

$$\text{threshold} = \max(0.05 \text{ g}, 4 \times \sqrt{(2 \cdot (\sigma/\sqrt{10})^2 + 2 \cdot p^2)})$$

Example, compartment 1: $\sigma/\sqrt{10} = 0.02 / 3.16 = 0.0063 \text{ g}$. The combined error is $\sqrt{(2 \times 0.0063^2 + 2 \times 0.02^2)} = 0.0297 \text{ g}$. Times 4 gives **0.119 g**.

Compartment	Threshold	A normal use	Margin
1 · Sachets	0.119 g	one sachet, 1.1 to 1.9 g	9 to 16 times
2 · Blisters	0.116 g	one pill, 0.13 to 0.37 g	1.1 to 3.2 times (tight, see failure list)
3 · Packages	0.184 g	one red ginseng stick, about 12 g	65 times
4 · Bottles	0.184 g	one capsule, 1.1 to 1.5 g	6 to 8 times

What 4 times means: if the noise behaves normally, an opening where nothing came out reads above the threshold about 1 time in 31,600.

Step 7. Tell an opening from a knock

An opening counts only if the weight changed by more than the threshold, or a hand was in the compartment for at least 1.5 seconds in total. A knock on the table lasts 0.1 to 0.3 seconds and leaves no weight change, so it is ignored. In the simulation test, 500 knocks produced 0 openings.

Step 8. Decide which dose time an opening counts for

Each dose time has a window. It opens 60 minutes early. It closes 180 minutes late, or halfway to the next dose time that needs the same compartment, whichever comes first. An opening counts for the nearest open window that is not already done.

The example schedule: morning 08:30 needs all four. Lunch 12:30 needs 1. Evening 18:30 needs 1, 2 and 4. Bedtime 22:00 needs 3 and 4.

Compartment	Morning 08:30	Lunch 12:30	Evening 18:30	Bedtime 22:00
1 · Sachets	07:30–10:30	11:30–15:30	17:30–21:30	not needed
2 · Blisters	07:30–11:30	not needed	17:30–21:30	not needed
3 · Packages	07:30–11:30	not needed	not needed	21:00–01:00
4 · Bottles	07:30–11:30	not needed	17:30–20:15	21:00–01:00

Worked example, compartment 1, morning: the next need is lunch at 12:30, which is 240 minutes later. Half of 240 is 120, which is less than 180, so the window closes at 08:30 + 120 min = 10:30. This follows the pharmacist's rule: "If it is nearly time for the next dose, skip the missed one."

Worked example, compartment 4, evening: the next need is bedtime at 22:00, 210 minutes later. Half is 105 minutes, so the window closes at 20:15.

Step 9. Remind in steps, only when someone is home

Time after the dose time	What Lyra does (person at home)
0 min	The compartment lights glow and the tiles turn black
+15 min	One soft chime
+30 min	One short voice line
+40, +50, +60 min	The voice line again, at most 3 more times
After that	Lights only, until the window closes

If the radar sees nobody, Lyra makes no sound. If the person comes back more than 15 minutes after the dose time, it speaks once. A compartment marked "quiet" (supplements) never goes past one chime and never alerts family.

Step 10. Tell a bottle lift from a dose

If more weight leaves than the compartment's largest normal use (3, 1, 13 and 2 g), a container is in the person's hand.

Example, compartment 4: the omega-3 bottle is lifted, net +182.4 g. That is more than 2 g, so the bottle is out. It comes back within 10 minutes, net -181.1 g. $182.4 - 181.1 = 1.3$ g, which is more than 0.05 g. So the record is "opened, something came out". If it had come back at the same weight, the record would be "opened, nothing came out".

If the bottle is still out after 10 minutes, the screen asks to put it back (screen 9), once.

Step 11. Decide "missed" and tell the family once

A needed compartment that is still unopened when its window closes is "missed". Lyra notes whether the radar saw the person at home during the window.

Example: compartment 2's evening window closes at 21:30. At 21:31 it is marked missed, "at home". Ji-eun gets one KakaoTalk message for that dose time, even if more than one compartment was missed.

Step 12. Say "take it with you"

When the radar changes from "someone here" to "nobody here", Lyra checks every dose time from 60 minutes before it to 180 minutes after it. If a needed compartment is still unopened, it speaks once for that dose time.

Example: at 11:50 the person leaves. Lunch is at 12:30, 40 minutes away, which is less than 60, and compartment 1 is unopened. Screen 8 appears with one voice line. Leaving again at 12:05 says nothing.

Step 13. Tell a refill from a dose

Weight going **in** with no container out is a refill, not an opening. Holding the hidden button for 3 seconds starts refill mode, and nothing is counted until it ends.

Step 14. Show fine dust in four steps

Lyra uses the Korean grades. It shows the worse of PM10 and PM2.5 as one word and a 4-dot scale.

Grade	PM10 ($\mu\text{g}/\text{m}^3$)	PM2.5 ($\mu\text{g}/\text{m}^3$)	Dots
좋음 (good)	0–30	0–15	1
보통 (moderate)	31–80	16–35	2
나쁨 (bad)	81–150	36–75	3
매우나쁨 (very bad)	151 and up	76 and up	4, with "mask when you go out"

Example: PM10 is 62 and PM2.5 is 41. PM10 alone is moderate, but PM2.5 is bad, so the screen shows 나쁨 with 3 dots.

Step 15. The weekly number the family sees

On-time rate = needed openings that happened inside their window $\div$ needed openings.

Example: the schedule above has $4 + 1 + 3 + 2 = 10$ needed openings a day, so 70 a week. If 67 happened in their windows, the rate is $67 \div 70 = 95.7\%$. It counts compartments opened on time. It does not count pills.

Every failure scenario

We listed 34 ways Lyra could fail or be fooled. Today 11 are tested in simulation, 8 are written in the firmware but not yet tested, and 10 are designed but not yet built. The other 5 are things Lyra cannot detect, and we say so openly.

Scenario	What Lyra does	Status
Sensing		
A knock on the table or on a compartment	Ignored. A knock lasts 0.1 to 0.3 s and leaves no weight change. 500 of 500 knocks were ignored.	Tested in simulation
Opened and looked, nothing taken	Logged as "opened, nothing came out" if a hand was inside for 1.5 s or more.	In code, not yet tested
Same compartment opened twice in one dose time	The second one is logged "opened again". No alarm, and no second dose is counted.	Tested in simulation
Opened when nothing is due	Screen 7 shows when it is next due. Logged, with no alarm.	Tested in simulation
Bottle or tube lifted out, used and put back	Counted as one opening. The weight on return decides "something came out".	Tested in simulation
Container left out	After 10 minutes, screen 9 and one voice line.	Tested in simulation
Container comes back after more than 10 minutes	Still counted as a return, not a refill.	Tested in simulation
A 12 g ginseng stick taken away for good	Normal use, not "container left out".	Tested in simulation
Refill: medicine added	Weight going in is not an opening. Holding the hidden button for 3 seconds starts refill mode.	Tested in simulation
One light pill (0.13 g) pressed out inside compartment 2	"Opened" is certain. "Something came out" may read no, because 0.13 g is close to the 0.116 g threshold. Lifting the pack out and back gives a clear reading.	Cannot detect
Scale drift over hours (temperature)	While nobody touches it, the zero follows slow changes, 5% per settled reading.	In code, not yet tested
Lyra itself lifted or moved	Rule: if 3 or more compartments change within 2 s, ignore it.	Designed, not built
Schedule and time		
Two windows of one compartment overlap	Cannot happen. Each window closes halfway to the next dose time.	In code, not yet tested
Bedtime dose taken after midnight	Counts for the previous night's bedtime until the window closes at 01:00.	In code, not yet tested
Dose time missed while at home	Marked "missed, at home". One family message.	Tested in simulation
Dose time missed while away	Marked "missed, away". The family message says the person was out.	In code, not yet tested
Several compartments missed at one dose time	One family message for the dose time, not one per compartment.	In code, not yet tested
Supplement compartment missed	Quiet compartment: light and one chime at most, never a family message.	Tested in simulation
Pharmacist changes the schedule	The family enters it on the setup web page. It applies from the next dose time.	Designed, not built

Scenario	What Lyra does	Status
Clock is wrong	Time comes from the internet (NTP) and a clock chip with its own battery.	Designed, not built
People and presence		
Leaves home before a dose time	"Take it with you" once for that dose time.	Tested in simulation
Comes home after the reminders	One voice line on return, if more than 15 minutes have passed.	In code, not yet tested
Radar misses a person sitting very still	Lyra skips the sound, but the lights still glow. A miss may be labelled "away". The family can set "sound always".	Designed, not built
Radar reacts to a fan, curtain or pet	A chime plays in an empty room. This is harmless.	Designed, not built
A visitor, carer or pet opens a compartment	It counts as an opening. Lyra cannot tell who opened it.	Cannot detect
The wrong sachet is torn from the roll	Lyra cannot see this. The dose time printed on each sachet is the safeguard.	Cannot detect
Medicine taken out but not swallowed	Lyra cannot see this.	Cannot detect
Two people share one Lyra	Their records would mix. The rule is one Lyra per person.	Cannot detect
Poor hearing	The lights and black tiles carry every message. Sound is optional.	Designed, not built
Poor eyesight	Each tile is about 39 × 25 mm with a picture of what is inside. Voice lines are in Korean. A large-view setting shows one compartment at a time (concept C).	Designed, not built
Home and power		
Wi-Fi is down	Everything works on the device. Messages wait in an outbox and go out when Wi-Fi returns.	In code, not yet tested
Power cut	E-ink keeps its image with no power. A backup battery keeps the clock and scales running, and the log is saved to flash.	Designed, not built
KakaoTalk message fails	It is retried from the outbox.	Designed, not built
Night	No sound between 23:00 and 07:00 unless a dose time is set then.	Designed, not built

The statuses will move to "Tested in simulation" and then to hardware tests as the work goes on.

The value, calculated

For one person on the example schedule, Lyra makes 3,650 time-stamped records a year. The family gets about 73 messages a year instead of 1,460. It is built from about \$52 of parts, with no required subscription.

For the person

- **No re-sorting.** The sachet roll, blisters, sticks and bottles go in as they are.
- **One place for everything.** Four compartments cover the four forms Korean medicine comes in.
- **An answer to "did I already take it?"** at any moment. The tile shows a check and the time.
- **A screen worth looking at every day.** It shows the clock, weather and fine dust, so the reminder sits where the eyes already go.

For the family

Item	Calculation	Result
Openings recorded per year	10 needed openings a day $\times$ 365	3,650
Dose times per year	4 a day $\times$ 365	1,460
Messages if every dose time sent one (typical app)	1 $\times$ 1,460	1,460 a year, 4 a day
Lyra messages if 1 dose time in 20 is missed	1,460 $\div$ 20	about 73 a year, 1.4 a week

The family hears from Lyra only when something needs them. So when a message comes, they read it.

Cost to own over 3 years (US prices)

Product	Calculation	3-year cost
Livi	$\$130 + 36 \times \99	\$3,694
Hero, month-to-month	$\$99.99 + 36 \times \44.99	\$1,719.63
Hero, 2-year prepaid rate	$\$99.99 + 36 \times \29.99	\$1,179.63
Lyra (proposed)	One-time price, no required subscription	To be set. Proposed target: under \$250

Competitor prices come from search summaries, linked under Sources at the end.

What Lyra costs to make and run (estimates)

These are approximate prices for 1,000 units, not yet quoted by suppliers.

Part	Estimate
4.2-inch e-ink panel	\$12.00
Enclosure and compartment trays	\$8.00
4 load cells and 4 HX711 amplifiers	\$8.00
Circuit board and assembly	\$6.00
ESP32-S3 module (16 MB flash, 8 MB RAM)	\$3.50
1,000 mAh backup battery and charger	\$3.50
USB-C power adapter	\$3.00

Part	Estimate
24 GHz radar (LD2410C)	\$2.50
Speaker and amplifier	\$2.00
Clock chip and coin cell	\$1.80
Temperature and humidity sensor (SHT40)	\$1.20
Hidden button and 4 lights	\$0.50
Total	about \$52

Electricity. Lyra draws about 0.8 W on average: Wi-Fi about 0.33 W, radar about 0.4 W, the rest about 0.07 W. $0.8 \text{ W} \times 24 \text{ h} \times 365 \text{ days} = 7.0 \text{ kWh}$ a year. That is about ₩1,000 to ₩2,000 a year at Korean home rates.

Power cut. The backup battery holds $1,000 \text{ mAh} \times 3.7 \text{ V} = 3.7 \text{ Wh}$. During a cut, Lyra turns off Wi-Fi and the radar and draws about 0.14 W. $3.7 \text{ Wh} \times 85\% \text{ usable} \div 0.14 \text{ W} = \text{about } 22 \text{ hours}$ of scales, clock and lights.

For the business

The global automatic pill dispenser market was USD 3.7 billion in 2025. It is forecast to reach USD 6.6 billion by 2034 ([IMARC](#), from a search summary). Almost all of it is pill-cup dispensers built for US pharmacy habits. Lyra is built for sachet-roll countries first: Korea first, and later Japan, where pharmacies also pack medicine by dose time (一包化).

What is new, and how it compares

Today's smart dispensers want your pills loose so that a machine can hand them out. Lyra does the opposite. It keeps medicine the way the pharmacy packed it, and it senses what happens to it.

Product	Korean sachet rolls	What the person must do first	Required subscription
Lyra	Yes, as they come	Put the medicine in its compartment	No
Hero	No, loose pills only	Load pills into its cartridges	Yes, \$29.99 to \$44.99 a month
MedMinder	No, pills in pre-sorted trays	Load a filled tray	Yes
Livi	No, loose pills only	Load pills into the machine	Yes, \$99 a month
Phone reminder apps	Any form, but they sense nothing	Use an app every day and tap "taken"	Varies

A Korean design study proposed an ATM-style sachet-dispensing machine for older users ([KCI](#)). Lyra reaches the same goal with no moving parts.

Eight things that are new together in Lyra:

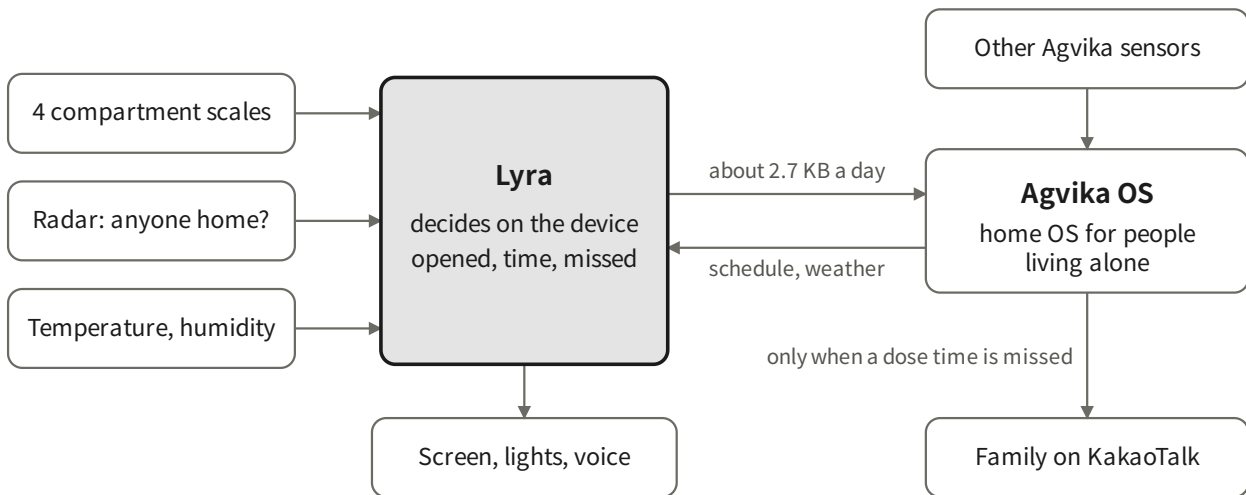
1. **Any medicine form.** Four scales cover sachet rolls, blisters, sticks, drops, tubes and bottles, with no repacking.
2. **The screen is a map of the box.** The screen draws the same 2×2 square as the compartments, with a picture of each form. A black tile means "open the one in this spot". No reading is needed.
3. **Honest records.** It records the compartment, the time and the weight change. It never claims "took the medicine".
4. **Reminders that know if you are home.** It stays silent when you are out, says "take it with you" as you leave, and speaks once when you return.
5. **Quiet for the family.** One message per missed dose time and nothing else, which is about 73 a year instead of 1,460.
6. **A reason to look every day.** The clock, weather and fine dust make Lyra part of the morning, not a medical box in a drawer.
7. **Nothing to install, nothing that watches.** There is no app for the user, no camera and no microphone.
8. **A daily-living sensor.** Openings at regular times are a daily sign of routine for Agvika OS (next section).

Design competitions in 2027. We plan to enter the CES Innovation Awards (Digital Health, and Accessibility and AgeTech), Red Dot, and the iF Design Award. The story for judges is short: *four tiles, four compartments, no reading.*

Connection to Agvika OS

Lyra sends Agvika OS about 2.7 KB of events a day, which is about 1 MB a year. The events carry compartment numbers, times and weight changes. They never carry sound, images or medicine names.

How Lyra connects to Agvika OS



Lyra makes every decision on the device, so it keeps working without the internet. Agvika OS receives the results and sends back the schedule, weather and fine dust.

What is sent (designed message format)

Message	Sent when	Example	Size	Per day
open	Each opening	<code>{"t":"open","ts":1790383260,"c":1,"s":0,"took":1,"g":1.52,"late":11}</code>	67 B	about 10, 675 B
back	A container is put back	<code>{"t":"back","ts":1790383300,"c":4,"g":-181.1}</code>	45 B	about 5, 225 B
dose	A dose time closes	<code>{"t":"dose","d":20727,"s":0,"need":[1,2,3,4],"done":[1,2,3,4],"home":1}</code>	64 B	4, 256 B
hour	Every hour	<code>{"t":"hour","d":20727,"h":14,"pres":38,"temp":24.1,"rh":46}</code>	58 B	24, 1,406 B
day	End of the day	First seen, last seen, on-time openings	77 B	1, 77 B
note	Leaving, container out, refill	<code>{"t":"note","ts":1790383260,"k":"leaving","s":1}</code>	48 B	about 1, 48 B
Total			about 2.7 KB	

The fields mean: **t** type, **ts** time, **c** compartment, **s** dose time (0 morning to 3 bedtime), **took** something came out, **g** grams, **late** minutes after the dose time, **pres** minutes someone was near Lyra in that hour. Messages travel as JSON over MQTT with TLS encryption. If Wi-Fi is down, they wait in an outbox.

What Agvika OS can learn from it

- **Daily routine.** Lyra reports when the person was first seen and when they first opened a compartment. A morning with no opening and no presence by the usual time is a strong sign that something is wrong. Agvika OS decides what to do using its own rules and its other sensors.
- **Activity.** Minutes of presence per hour show the person's daily rhythm near the kitchen or living room.

- **Room climate.** Hourly temperature and humidity can warn of heat in summer or cold in winter for someone who lives alone.
- **Medication routine.** The weekly on-time rate shows if the routine is slipping, weeks before a hospital visit would.

Privacy

- Lyra has no camera and no microphone. The radar senses movement, not images.
- Compartments are numbers. Lyra never knows medicine names.
- We treat the data as sensitive health information under Korea's Personal Information Protection Act. We ask for separate consent at setup.
- The person can see what Lyra recorded today with one press of the hidden button.
- The family sees only missed dose times and a weekly summary.

How Lyra gets smarter every day

Lyra learns the person's own rhythm from their openings and presence. It changes *when and how* it reminds, and never *what* the medicine schedule is. The first version uses simple statistics on the device. A language model is used only to write the weekly family message, from numbers alone.

What Lyra knows as the days go by

Day	What Lyra knows	What changes for the person
Day 1	Only the schedule	Standard reminders: light at the dose time, chime at +15, voice at +30
Day 14	The usual opening time for each dose time	Reminders come only when the person is later than usual
Day 28	Weekday habits and the normal range of each time	"Take it with you" comes before the usual outings. The family summary can spot changes.
Day 90	The seasonal rhythm, together with Agvika OS	Agvika OS can tell an unusual day from a normal one

What it learns, with examples

- Reminders at the person's own time.** Lyra keeps the median opening time of each dose time over the last 14 days. The chime moves toward it, at most 30 minutes earlier or 45 minutes later. The window never moves. *Example:* the morning dose is at 08:30, but the person always opens it after breakfast, at a median of 08:55. The chime moves from 08:45 to 09:10, so there is no chime on normal days. The window still closes at 10:30.
- Routine drift.** Each week's median opening time is compared with the 4 weeks before. A change of more than 45 minutes goes into the family summary. *Example:* bedtime openings moved from 22:05 to 23:10 this week.
- Weekday patterns.** A dose time missed on the same weekday for 3 of the last 4 weeks is a pattern. *Example:* Sunday lunch is missed while the person is out. The family summary suggests packing a sachet on Sundays.
- Usual outings.** Presence history shows regular absences. *Example:* every Tuesday the person is out from 10:30 to 15:00 at the senior centre. So at 10:00 on Tuesdays, Lyra says to take the lunch sachet along.
- The right reminder strength.** If the person always responds to the light, the chime is rarely needed. If a dose time often needs the voice, the voice comes 15 minutes earlier at that dose time only.
- A weekly family message written in Korean.** The code counts the numbers. A language model turns those numbers into two or three warm sentences. It never sees raw data, and it cannot change the numbers.

Example weekly message: "이번 주 약 칸 열기: 70번 중 67번 제때 여셨어요. 일요일 점심은 외출 중에 놓치셨어요 (3주째). 저녁 시간이 평소보다 40분 늦어졌어요." This means: "This week she opened the compartments on time 67 of 70 times. She missed Sunday lunch while out (third week). Evenings are 40 minutes later than usual."

Results shown on the screen

Lyra shows an AI result only when it asks for one clear action. Three do (screens 12, 13 and 14 in "Every screen").

Days of sachets left (screen 12). A sachet roll is nothing but sachets, so its weight falls by the same amount every day.

- Weight in compartment 1 now, minus the empty tray: 13.4 g.
- Average daily drop over the last 3 days: $(4.4 + 4.5 + 4.5) \div 3 = 4.47$ g a day. That matches one day's sachets: $1.85 \text{ g} + 1.10 \text{ g} + 1.50 \text{ g} = 4.45 \text{ g}$.

3. Days left = $13.4 \div 4.47 = 3.0$ days, so Saturday, Sunday and Monday.
4. At 3 days or fewer, the screen shows the pharmacy day: Monday.

This works for sachets because the roll has no container. For blisters and bottles, the empty pack also has weight, so the family would enter the empty weight at setup.

A learned outing (screen 13). For each weekday, Lyra keeps the hours when the radar saw nobody. If the person was away across a dose time on 3 of the last 4 same weekdays, that is a usual outing. *Example:* away from 10:30 to 15:00 on 3 of the last 4 Tuesdays, and lunch at 12:30 falls inside. So at 10:00, 30 minutes before the usual departure, the screen says to pack the lunch medicine.

This week (screen 14). Needed openings on time $\div$ needed openings = $67 \div 70$. It shows on Sunday morning, the first time the person comes near.

Now and next

Capability	Where it runs	When
Reminder at the person's own time	On the device	Version 1
Routine drift and weekday patterns	On the device	Version 1 (similar logic is already tested in our earlier model)
Weekly family message in Korean	Cloud, numbers only	Version 1
Usual outings	On the device	Version 2
Right reminder strength	On the device	Version 2
Unusual-day detection with other sensors	Agvika OS	Version 2, with the Agvika OS team
Days of sachets left, on the screen	On the device	Version 1
Weekly result on the screen	On the device	Version 1

What AI never does: change a dose, tell the person to take or skip a medicine, or make a medical judgement. Those stay with the doctor and pharmacist.

Evidence today, and what is not proven yet

The logic is written and passes its simulation tests. The screens are designed as vector files. Nothing has been measured on hardware yet, and we mark every estimate as one.

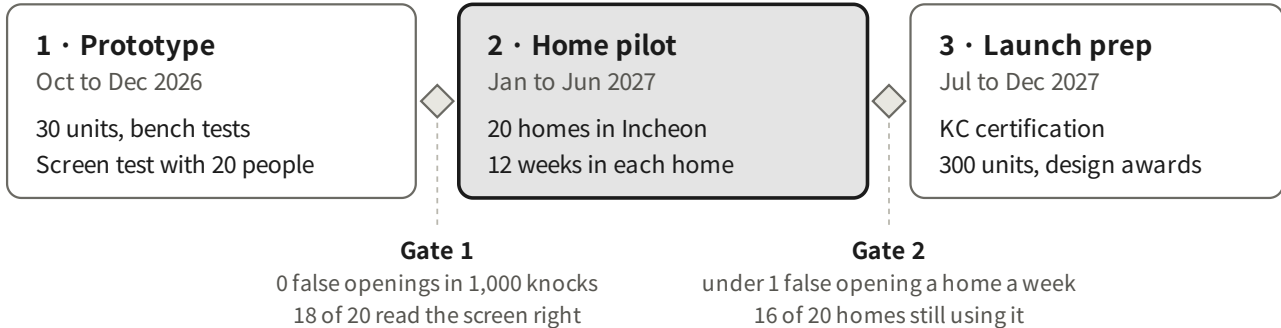
Part	Status	Evidence
Opening detection (scales, knocks, sessions)	Written in C++, tested in simulation	One simulated day from raw 10-per-second scale readings passes on 200 random seeds. 500 knocks gave 0 false openings.
Dose-time logic (windows, missed, family message, not due, again)	Written in C++, tested in simulation	The same day test checks every opening, the missed evening dose and exactly one family message.
Refill, container out, leaving home	Written in C++, tested in simulation	8 checks pass, including a bottle returned after 16 minutes
Screen design (16 screens, 2 other concepts)	Designed	Vector SVG and PNG files, and one UI kit artboard
AI features (own reminder time, drift, days left, outings)	Designed	Calculations in this document. Similar drift and pattern logic was tested in our earlier model.
ESP32-S3 device software	Drafted, not compiled	The build tools could not be downloaded in our work environment
Hardware: scale noise, radar range, battery life	Not measured	All such figures in this document are estimates
Older people understanding the screen	Not tested	Planned as the first gate in the next section

The simulation code is in the project repository (`lyra/firmware`), and anyone can run it with one command: `make -C lyra/firmware/test/host run`.

The plan with Incheon Techno Park, and our ask

We propose three phases over 15 months. A 20-home pilot in Incheon sits at the centre, and each phase must pass a measurable gate before the next starts. The dates and targets are our proposal, to agree together.

Proposed plan with Incheon Techno Park



Gate 1 tests the two riskiest claims first: that the scales never invent an opening, and that older people understand the 2×2 map in 3 seconds with no training.

What we ask from Incheon Techno Park (proposed):

1. **A 20-home pilot.** Introductions to senior care services in Incheon that visit older people living alone, so we can recruit 20 homes for 12 weeks.
2. **Prototype support.** Access to prototyping and test equipment to build and calibrate 30 units.
3. **Certification guidance.** Help with KC certification for the Wi-Fi and radar parts.
4. **Introductions.** Pharmacies and clinics in Incheon that dispense sachet rolls, and investors in care technology.

What Incheon gets: a Korean-designed, export-ready care device tested first in Incheon homes. It also gets pilot data (anonymous, with consent) on how older people living alone keep their medicine routine.

See it:

- Design files (vector SVG for Illustrator, and PNG): [lyra/design in the repository](#).
- Logic and tests: [pull request](#)

Appendix: the UI system

The same parts as the UI kit file: the map, the form icons, the tile states and the rules for what shows when.

1 The map

Lyra, top view
small circle = the hidden button, under the front-right edge

Thescreen: same 2 x 2
Black tile = open the compartment in that spot

2 Form icons

1 · 약봉투
sachet roll

2 · 블리스터
blister pack

3 · 스틱·안약·연고
stick, drops, tube

4 · 영양제 병
bottle

아침 점심 저녁 자기전

fine dust, 4 steps

3 Tile states

Due: open this one

Opened, with time

Not needed now

Dose time ended

Put it back here

Opened when not due

4 What the screen shows (most important first)

- 1 Container out 10 min → put it back (9)
- 2 Person leaves near a dose time → take it with you (8)
- 3 A dose time is open → the 2 x 2 map, black = open (4, 5)
- 4 AI: a usual outing is due → pack the next dose (13)
- 5 AI: 3 days of sachets or fewer → pharmacy day (12)
- 6 A dose time just ended with one unopened → slash, family told (10)
- 7 Someone within 1.5 m, nothing due → next dose time (3)
- 8 Nobody near → clock, weather, fine dust (1, 2, 11)

Hidden button: short press = today (15). Hold 3 s = refill mode (16).
E-ink: page change = full refresh; a tile flip = quiet partial refresh.
Type: Noto Sans KR 900 / 700 / 500 · numbers Schibsted Grotesk 800 / 700
Colour: ink #1B1B1B · paper #E8E7E1 · grey #8F8D86 (a 50% dither on the panel)

Files: every screen as SVG (vector, opens in Adobe Illustrator) and PNG, the UI kit as one SVG artboard, and the two diagrams as SVG. Fonts: Noto Sans KR and Schibsted Grotesk (both free from Google Fonts). Outlined copies need no fonts.

Sources

The network in our work environment blocked these pages, so the figures come from search result summaries. Please check them before sending.

- [Korea becomes a super-aged society \(Korea Times, MOIS data\)](#)
- [2025 고령자 통계 \(Statistics Korea\)](#)
- [Prevalence and Predictors of Polypharmacy among Korean Elderly \(PLOS ONE, 2014\)](#)
- [Hero pill dispenser review and pricing \(The Senior List\)](#)
- [Automatic pill dispensers compared, including Livi and MedMinder \(Caring Village\)](#)
- [Automatic pill dispenser machine market \(IMARC\)](#)
- [A user-centered smart medication dispenser design for Korean elderly users \(KCI\)](#)